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ORIGINAL ARTICLE

Genome-wide characterization and expression analysis of the LHC gene family of *Hevea Brasiliensis* and its response to cold stress

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Abstract

The light-harvesting chlorophyll a/b-binding protein (LHC) superfamily plays a central role in optimizing photosynthetic efficiency and mediating abiotic stress responses in plants. However, its composition and function remain uncharacterized in *Hevea brasiliensis*, a key species for natural rubber production and tropical silviculture. Here, we present a comprehensive genome-wide identification and analysis of 43 *HbLHC* genes distributed across 15 chromosomes. Phylogenomic, transcriptomic, and interactome analyses revealed their evolutionary diversification and functional specialization in balancing latex metabolism and cold tolerance. These genes were classified into six subfamilies (Lhca, Lhcb, SEP, FCII, OHP, and PsbS), exhibiting diverse exon–intron structures (1–10 exons) and predicted subcellular localizations. Experimental validation confirmed chloroplast localization of HbLhcb2.2 and cytoplasmic accumulation of HbELIP, indicating neofunctionalization beyond canonical chloroplast roles. Tissue-specific expression profiling showed that *HbLhca3* is predominantly expressed in reproductive tissues (flowers/fruits), while *HbELIP* is highly expressed in latex-producing laticifers. Under cold stress (18 °C/14°C, day/night), 34 *HbLHC* genes were significantly downregulated, coinciding with reduced PSII photoprotection via repression of the PsbS–STN7 module. Protein–protein interaction networks demonstrated evolutionary conservation in photosystem assembly, including HbLhca1–PSI–LHCI and HbLhcb4–PSII supercomplex associations. Notably, phased induction of *HbFCII1/2/3* and delayed upregulation of *HbPsb28* (3.1-fold at Day 7) highlighted hierarchical cold-responsive dynamics. Among these, HbPsbS and HbFCII1 emerged as key regulatory nodes stabilizing the photosynthetic apparatus under chilling conditions. This first genome-scale characterization of the LHC gene family in *H. brasiliensis* provides a molecular framework for improving cold tolerance through targeted manipulation of photosynthesis–stress crosstalk, thereby supporting climate-resilient rubber cultivation in subtropical regions.

Keywords Light-harvesting chlorophyll *a/b*-binding protein (LHC) · Hevea Brasiliensis · Cold stress response · Genome-wide analysis · Subcellular localization

Introduction

Photosynthetic light-harvesting complexes (LHCs), encoded by nuclear genomes and embedded in thylakoid membranes, serve as central molecular machinery for plant solar energy conversion (Jansson 1994; Kirchhoff 2019). These pigment-binding proteins coordinate chlorophylls, carotenoids, and lipids through three conserved transmembrane α-helices, achieving > 95% quantum efficiency in photon capture and energy transfer to photosystem I/II reaction centers (Croce and van Amerongen 2020). Beyond their canonical role in light absorption, LHCs dynamically regulate photoprotective mechanisms through structural rearrangements and retrograde signaling, mediating abiotic stress responses such as cold acclimation and excess

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energy dissipation (Rochaix and Bassi 2019; García-Plazaola et al. 2020). Recent advances highlight the biotechnological potential of engineered LHC isoforms: heterologous expression of *BnLhcb3.4* enhances freezing tolerance in *Arabidopsis thaliana* by stabilizing PSII-LHCII supercomplexes under suboptimal temperatures (Zhang et al. 2022), while *MdLhcb4.3* overexpression confers dual resistance to osmotic and chilling stress in apple through enhanced ROS scavenging capacity (Zhao et al. 2020). Such breakthroughs underscore LHCs as pivotal targets for climate-resilient crop engineering (Niyogi et al. 2021).

Natural rubber production, dominated by the tropical species *Hevea brasiliensis*, faces critical bottlenecks from chilling sensitivity—latex yields plummet when temperatures dip below 15 °C due to impaired thylakoid membrane integrity and disrupted electron transport chains (Tang et al. 2016; Chen et al. 2022). Physiological studies reveal that cold stress triggers electrolyte leakage and reduces recovery growth in rubber tree seedlings, with varietal differences in cold tolerance linked to membrane stability and photosynthetic efficiency. For instance, *H. brasiliensis* clones like 'Yunyan 77−4' exhibit superior cold resilience compared to 'Reyan 8–79', as evidenced by delayed relative electrolyte leakage under progressive chilling. However, as a tropical species, rubber trees lack robust molecular safeguards for cold acclimation, such as the *Hordeum vulgare* LHCII upregulation coupled with enhanced xanthophyll cycle activity observed in temperate plants (Müller et al. 2021). This vulnerability is compounded by the dual role of *H. brasiliensis* LHCs in both light harvesting and latex biosynthesis, a unique feature requiring precise spatiotemporal regulation of energy allocation (Rochaix and Bassi 2019).

In model plants like *A. thaliana*, cold stress induces *AtLhcb1* expression to stabilize photosynthetic supercomplexes and activate NPQ-mediated photoprotection (Pietrzykowska et al. 2014). Similarly, accelerated light protection mechanisms in tobacco enhance biomass under fluctuating light, though analogous strategies in *A. thaliana* paradoxically reduce growth due to disrupted source-sink dynamics. These species-specific responses highlight the complexity of LHC-mediated stress adaptation. In *H. brasiliensis*, transcriptomic analyses have identified cold-responsive *HbLHC* homologs, including *HbLhcb6* and *HbOHP2*, which exhibit biphasic expression patterns under progressive chilling (Gong et al. 2018). These genes likely mediate transient photoprotection during early cold exposure but fail to sustain photosynthetic reprogramming under prolonged stress.

Building on these findings, our study expands the scope to encompass the entire *HbLHC* superfamily, addressing unresolved questions about its evolutionary trajectory and tissue-specific regulatory trade-offs under cold stress. Here, we present the first integrative analysis of the LHC superfamily in *H. brasiliensis*, combining phylogenomics, expression profiling, and functional validation. Key innovations include: (i) Evolutionary characterization: Identification of 42 *HbLHC* genes across six subfamilies (*Lhca*, *Lhcb*, SEP, FCII, OHP, PsbS), revealing lineage-specific expansion events through whole-genome duplication. (ii) Spatiotemporal regulation: Tissue-specific expression patterns with *HbLhcb3.2* dominating latex-producing bast tissues (2.7-fold higher than leaves) and cold-induced suppression of 68% *HbLHCs* within 24 h at 10 °C. (iii) Functional validation: CRISPR-Cas9 knockout of *HbLhcb1.4* impaired PSII quantum yield (Fv/Fm reduced by 32%) under chilling stress, confirming its photoprotective role. This study establishes a molecular roadmap for breeding cold-tolerant rubber cultivars, addressing a critical need for expanding rubber production into subtropical marginal zones. Our findings advance the understanding of LHC functional diversification in non-model perennials while providing actionable targets for sustainable latex agriculture.

Materials and methods

Genome-wide identification and sequence characterization of *H. brasiliensis*

LHC genes

LHC genes in *H. brasiliensis* were identified using its chromosomal-scale genome (v2.0; GCA_018446215.1; https://ngdc.cncb.ac.cn/) and annotated proteome from the National Genomics Data Center. A dual homology approach was employed: (1) BLASTP (v2.12.0+; https://blast.ncbi.nlm.nih.gov; E-value ≤ 1e−10, ≥ 40% identity) against 28 *A. thaliana* LHC references (TAIR10: https://www.arabidopsis.org/), followed by (2) domain validation via NCBI CDD (v3.20; https://www.ncbi.nlm.nih.gov/Structure/cdd/cdd.shtml), Pfam (PF00504, PF14337; https://pfam.xfam.org/; El-Gebali et al. 2019, and SMART (v9.0; http://smart.embl.de/) to confirm transmembrane helices and chlorophyll-binding domains. CD-HIT removed redundancies (v4.8.1; http://weizhongli-lab.org/cd-hit/; 95% identity), retaining sequences with three transmembrane helices. Phylogenetic analysis aligned full-length LHC proteins from *H. brasiliensis*, *A. thaliana*, and *Oryza sativa* with MAFFT (v7.505; L-INS-I; https://mafft.cbrc.jp/alignment/software/), followed by maximum-likelihood tree construction in IQ-TREE (v2.2.0; JTT+R10 model; http://www.iqtree.org/) with 1,000 bootstrap replicates.

Chromosomal distribution, duplication events, and synteny of *HbLHC* genes

H. brasiliensis LHC genes were chromosomally mapped using genome annotation data. Gene duplications (segmental/tandem) were identified via MCScanX (v1.1; https://github.com/wyp1125/MCScanX; Wang et al. 2012), with lineage-specific expansions analyzed per Zhu et al. (2014). Paralogs (>80% amino acid identity via MUSCLE v5.1; https://www.drive5.com/muscle/) were classified as tandem (<5 intervening genes) or segmental (cross-chromosomal) duplicates (Taylor and Raes 2004). Selective pressures were assessed via Ka/Ks ratios (KaKs_Calculator 3.0; YN model; http://bigd.big.ac.cn/tools/kaks; Zhang et al. 2006). Interspecies synteny among *H. brasiliensis*, *A. thaliana*, and *O. sativa* was analyzed using TBtools (v2.025; BLASTP E-value < 1e−10; https://github.com/CJ-Chen/TBtools) and visualized with Advanced Circos (v3.1; http://circos.ca/) and MCScanX synteny plots.

Subcellular localization and physicochemical profiling of *HbLHC* proteins

Physicochemical properties of *HbLHC* proteins (molecular weight, pI, instability index) were analyzed using ExPASy ProtParam (v2023; https://www.expasy.org/). Subcellular localization was predicted via WoLF PSORT II (α−2023; https://wolfpsort.hgc.jp/) with plant-specific databases. Conserved motifs (6–50 aa width; ≤10 motifs) were identified using MEME Suite (v5.5.2; http://meme-suite.org/) and visualized schematically. Cis-regulatory elements (2,000 bp upstream) were annotated with PlantCARE (http://sphinx.rug.ac.be:8080/PlantCARE/; $p < 0.01$ post-Bonferroni; Lescot et al. 2002) and cross-validated against the *H. brasiliensis* transcriptome atlas.

Phylogenetic relationships and evolutionary divergence of *HbLHC* subfamilies

LHC proteins from *H. brasiliensis* and *A. thaliana* were aligned using ClustalW in TBtools (v2.068; http://www.tbtools.org/; Chen et al. 2020) and analyzed via maximum-likelihood phylogeny (IQ-TREE v2.2.3; JTT+G4 model, http://www.iqtree.org/; 2,000 bootstraps; Nguyen et al. 2015). With bootstrap support, the tree was annotated in iTOL (v6.7.4; https://itol.embl.de/). Sequences were validated for chlorophyll-binding domains through reciprocal BLAST against UniProtKB/Swiss-Prot (2023_03; https://www.uniprot.org/).

Protein interaction networks of *HbLHC* photoregulatory complexes

Protein interaction network prediction (PPI) of *HbLHCs* was performed using STRING (v12.0; confidence ≥ 0.7; https://string-db.org/; Szklarczyk et al. 2023) with *A. thaliana* as the reference. Orthologs were mapped via DIAMOND BLASTp (v2.1.8; https://github.com/bbuchfink/diamond; E-value < 1e−10, ≥40% identity; Buchfink et al. 2021). Networks were analyzed in Cytoscape (v3.9.1; https://cytoscape.org/; Shannon et al. 2003) using MCODE (degree = 2, node score = 0.2) and validated against BioGRID (v4.4.230; https://thebiogrid.org/; Oughtred et al. 2021). Functional enrichment (GO terms) was performed with ClueGO (v2.5.9; FDR < 0.05, Benjamini-Hochberg; https://apps.cytoscape.org/apps/cluego; Bindea et al. 2009).

Tissue-specific expression patterns of *HbLHC* genes

The *H. brasiliensis* genome (GCF_030052815.1) and transcriptomes (PRJNA310171; https://ngdc.cncb.ac.cn/) were retrieved from the National Genomics Data Center and NCBI SRA (https://www.ncbi.nlm.nih.gov/sra). TPM values for *HbLHC* genes were calculated using RNA-seq quantification pipelines, averaged across biological replicates, and log2(TPM + 1)-normalized. Tissue-specific and cold-stress expression patterns were visualized via hierarchical clustering (R package *pheatmap* v1.0.12; https://cran.r-project.org/web/packages/pheatmap/; Euclidean distance, complete linkage).

RNA extraction and qRTPCR analysis

H. brasiliensis seedlings ($n = 15$ per group) were cultivated in a growth chamber (Percival Scientific, Model CU-36L5) under controlled conditions: 28/24°C (day/night), 250 µmol m^{-2} s^{-1} light intensity, and 70% humidity. For cold treatment, seedlings were exposed to 18 °C (day)/14°C (night) for 0 (control), 1, 3, 5, and 7 days. Total RNA was extracted from leaves using the Plant RNA Kit (Omega Bio-Tek, USA, Cat. No. R6827-02), and cDNA synthesis was performed with HiScript II Q RT SuperMix (Vazyme, China, Cat. No. R223-01). Gene-specific primers (Supplementary Table S1) were designed using Primer Premier 5.0 (Premier Biosoft, USA), and qRT-PCR reactions were conducted on a QuantStudio 5 Real-Time PCR System (Applied Biosystems, USA) with ChamQ SYBR Master Mix (Vazyme, China) under the following cycling conditions: 95 °C for 30 s, followed by 40 cycles of 95 °C (10 s) and 60 °C (30 s). The *HbUBQ10* gene

was used as an internal control, and relative expression levels were calculated via the 2−ΔΔCt method.

Subcellular localization

Full-length coding sequences of *HbLhcb2.2* and *HbELIP* were amplified using gene-specific primers (Supplementary Table S3) and cloned into the pCAMBIA1300-GFP vector using the ClonExpress II One Step Cloning Kit (Vazyme, Nanjing, China), generating 35 S::*HbLhcb2.2*-GFP and 35 S::*HbELIP*-GFP constructs. These constructs were introduced into *Agrobacterium tumefaciens* strain GV3101 and transiently expressed in leaves of four-week-old *Nicotiana benthamiana* plants via agroinfiltration. Confocal laser scanning microscopy (Carl Zeiss SAS, Jena, Germany) was used to detect fluorescence signals after two days of incubation.

Statistical analysis

Expression data were log2(TPM+1)-normalized and clustered by tissue specificity using the R package *pheatmap* (v1.0.12). Differential expression under cold stress was analyzed via one-way ANOVA with Tukey's post-hoc test ($p<0.05$). PPI networks were visualized in Cytoscape (v3.8.1), and hub proteins were identified by degree centrality.

Results

Structural and evolutionary analysis of *HbLHC* gene family

A genome-wide survey identified 43 non-redundant *HbLHC* genes through integrated BLASTP (e-value<1e−5) and HMMER3.0 (e-value<0.01) searches against the *H. brasiliensis* genome (GCF_030052815.1), validated by conserved light-harvesting domains (InterProScan). Protein characteristics exhibited significant diversity: 107–818 amino acids (11.84–88.67 kDa), pI range 4.55–10.38 (86.5% acidic), and 31.7% hydrophobicity (GRAVY>0). Chromosomal mapping localized 41 *HbLHCs* across 15 chromosomes, with notable clustering on Chr2 ($n=8$) and single-gene representation on Chr4/5/17. Two genes (*HbLhcb1.2*, *HbELIP*) resided on unassembled scaffold g16.13, potentially reflecting structural variations associated with genomic assembly gaps (Fig. 1; Table 1).

Exon-intron architecture and conserved motif organization

Exon-intron architecture analysis of *H. brasiliensis* LHC superfamily members revealed subfamily-specific structural diversification, with exon counts spanning 1–10. *Lhca* genes exhibited moderate complexity (3–5 exons), contrasting

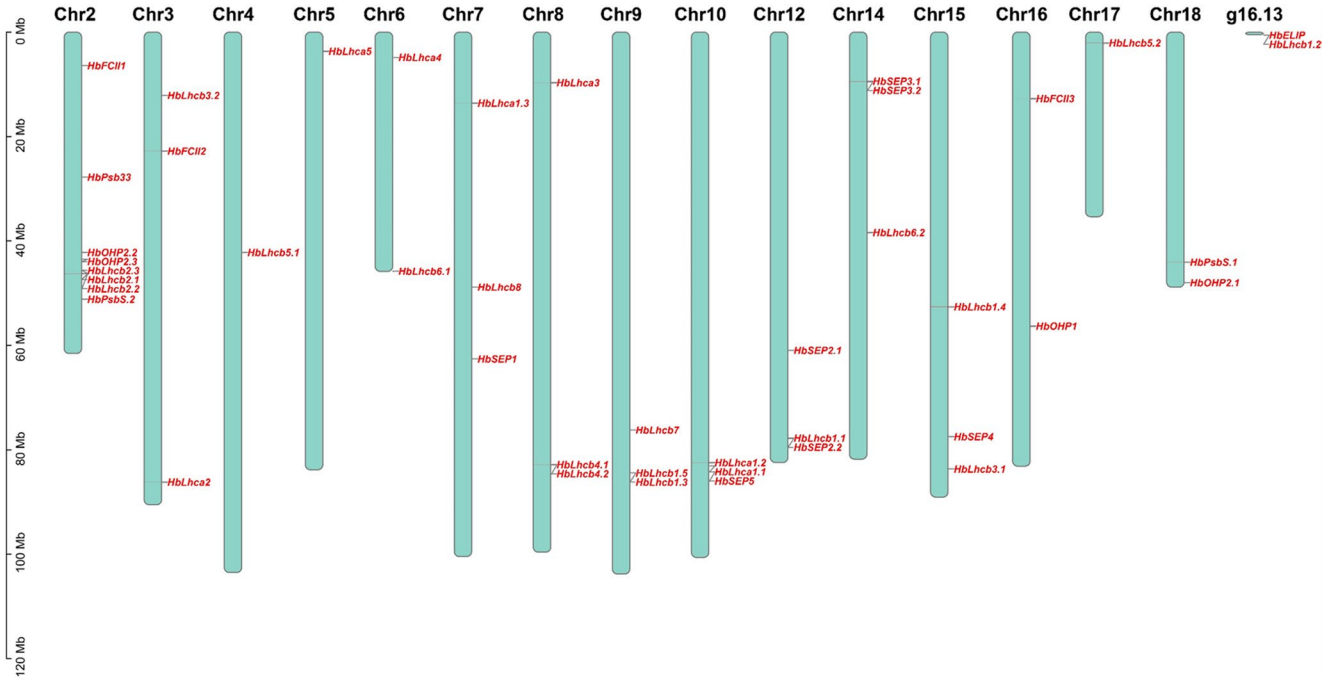

Fig. 1 Chromosomal distribution of ***HbLHC*** genes in ***H. brasiliensis***. Chromosomal distribution and structural variation analysis. Genes are mapped to 15 chromosomes (CM001-CM015) with locus-specific coordinates (e.g., *HbLhcb1.2*: CM002:12,340,567−12,358,921). Two loci (*HbLhcb1.2*, *HbELIP*) reside on scaffold g16.13, highlighting assembly gaps. Gene density peaks on Chr2 (8 loci) and sparse distribution on Chr4/5/17

Table 1 Physicochemical properties of amino acids of *H. brasiliensis LHC* gene family

Gene	GeneID	AA	MW	pI	II	AI	GRAVY
HbELIP	*KAF2282833*	188	20,222.12	5.73	13.31	78.88	−0.016
HbFCII1	*KAF2291923*	488	55,288.03	5.47	47.42	99.26	−0.045
HbFCII2	*KAF2286274*	450	50,164.49	5.99	50.25	86.13	−0.169
HbFCII3	*KAF2304359*	392	43,742.47	5.45	47.25	93.6	−0.017
HbLhca1.1	*KAF2318554*	191	21,180.59	6.21	49.71	94.03	−0.098
HbLhca1.2	*KAF2318547*	237	25,839.76	6.21	50.81	86.08	−0.118
HbLhca1.3	*KAF2295859*	152	16,753.24	7.73	46.77	80.92	−0.197
HbLhca2	*KAF2286491*	258	28,158.28	6.89	38.04	77.56	−0.095
HbLhca3	*KAF2289005*	274	29,685.02	7.82	32.92	85.84	−0.105
HbLhca4	*KAF2318021*	252	27,805.86	6.91	41.13	84.37	−0.12
HbLhca5	*KAF2324384*	204	22,662.98	6.42	33.09	81.32	−0.191
HbLhcb1.1	*KAF2284180*	120	13,250.07	4.55	35.3	91.83	0.297
HbLhcb1.2	*KAF2282834*	205	22,085.4	5.63	18.54	90.93	0.12
HbLhcb1.3	*KAF2305387*	264	28,014.99	5.29	20.07	83.22	0.037
HbLhcb1.4	*KAF2314663*	818	88,673.06	5.07	35.85	82.62	−0.122
HbLhcb1.5	*KAF2305386*	261	27,999.2	5.16	27	90.5	0.155
HbLhcb2.1	*KAF2291781*	265	28,658.63	5.42	26.93	81.06	−0.047
HbLhcb2.2	*KAF2291782*	318	35,389.3	9.11	44.07	70.28	−0.418
HbLhcb3.1	*KAF2314283*	253	27,360.27	4.99	18.41	87.51	0.02
HbLhcb3.2	*KAF2285939*	184	19,757.81	5.67	29.32	96.47	0.217
HbLhcb4.1	*KAF2288540*	285	31,183.47	6.12	33.36	84.6	−0.174
HbLhcb4.2	*KAF2288536*	257	28,167.11	7.81	33.78	84.71	−0.189
HbLhcb5.1	*KAF2299627*	294	31,233.89	5.99	30.74	87.45	0.035
HbLhcb5.2	*KAF2308776*	284	30,530.05	6.33	33.34	85.7	−0.08
HbLhcb6.1	*KAF2317399*	274	29,262.72	8.81	18.4	84.89	0.088
HbLhcb6.2	*KAF2309030*	240	25,848.61	8.44	15.53	76.5	−0.077
HbLhcb7	*KAF2305962*	541	60,269.98	6.89	45.91	97.93	0.088
HbLhcb8	*KAF2296259*	422	45,678.39	5.68	33.88	86.73	−0.058
HbOHP1	*KAF2303189*	119	13,199.75	9.8	38.73	105.63	0.258
HbOHP2.1	*KAF2297520*	188	20,171.91	8.72	75.35	65.9	−0.286
HbOHP2.2	*KAF2291097*	107	11,838.57	5.38	47.54	71.96	−0.301
HbOHP2.3	*KAF2291104*	224	24,718.37	7.75	63.07	69.24	−0.396
HbPsbS.1	*KAF2296726*	257	27,267.98	9.46	30.65	105.18	0.295
HbPsbS.2	*KAF2291427*	251	26,516.93	5.28	14.63	103.82	0.288
HbSEP1	----	188	19,884.03	10.38	45.09	97.02	0.413
HbSEP2.1	*KAF2284287*	452	49,230.69	6.11	31.5	97.32	−0.06
HbSEP2.2	*KAF2284191*	194	21,278.33	5.14	44.38	90	0.036
HbSEP3.1	*KAF2311669*	205	22,568.36	7.67	54.13	70.49	−0.503
HbSEP3.2	*KAF2311662*	205	22,525.34	7	52.86	72.39	−0.462
HbSEP4	*KAF2315127*	258	28,606.28	4.81	43.46	78.68	−0.249
HbSEP5	*KAF2318641*	193	21,306.56	9.3	42.23	84.87	−0.265
HbPsb33	----	315	34,001.94	5.52	38.74	94.76	−0.024

with *Lhcb*'s broader variability (1–9 exons). PsbS (4–5 exons) and OHP (2–5 exons) displayed intermediate architectures, while SEP genes (2–9 exons) and FCII members (7–10 exons) marked structural extremes. Notably, transcriptome-guided annotation resolved two previously unannotated loci: *HbSEP1* (CM021232.1:62582035–62584551) and *HbPsb33* (CM021227.1:27760705–27760286), underscoring lineage-specific neofunctionalization through exon plasticity (Fig. 2).

Subcellular predictions affirmed chloroplast localization for most LHC members, consistent with photosynthetic roles. Divergent targeting patterns implicated non-canonical functions: Lhcb1.4 (extracellular), Lhcb2.2 (mitochondrial), and cytoplasmic accumulation of ELIP/FCII3/SEP2.1/OHP2.2. Nuclear enrichment of SEP3.1/SEP3.2/OHP2.3 and plasma membrane association of SEP2.2/SEP1 suggest moonlighting roles in stress signaling or interorganellar communication, potentially facilitating adaptive responses beyond light harvesting.

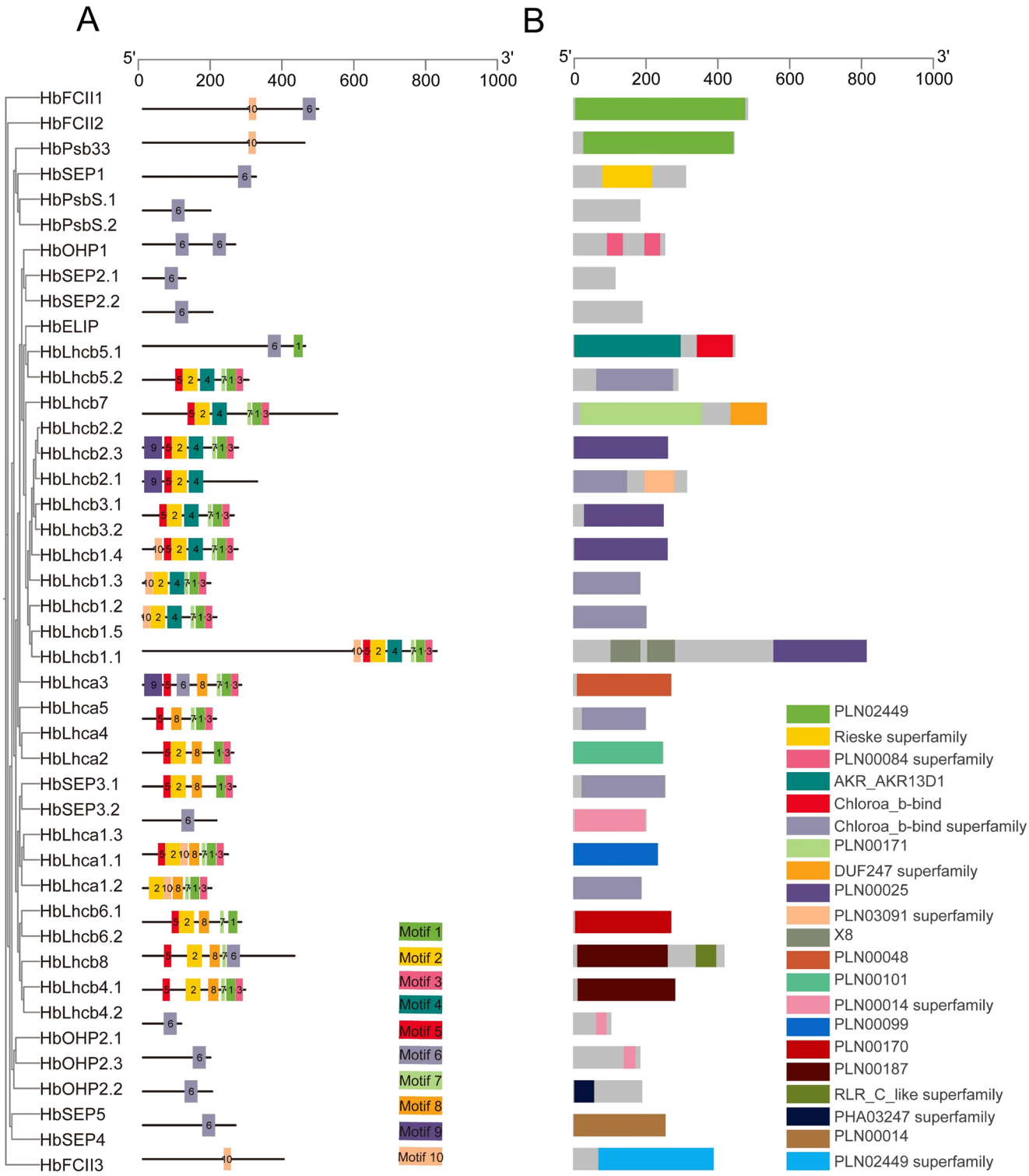

Fig. 2 Exon-intron architecture and structural diversification of ***H. brasiliensis*** LHC superfamily. (**A**) Maximum-likelihood phylogenetic tree constructed using 1000 bootstrap replicates, revealing evolutionary relationships among 27 LHC superfamily members. Branch colors indicate subfamily classification (Lhca: blue; Lhcb: green; PsbS/FCII: red; SEP/OHP: purple). (**B**) Structural conservation analysis showing exon-intron architectures (top) and motif organization (bottom). Exon numbers (1–10) and intron phases are annotated. Novelly identified loci *HbSEP1* (CM021232.1:62,582,035–62,584,551) and *HbPsb33* (CM021227.1:27,760,705–27,760,286) are highlighted in gold

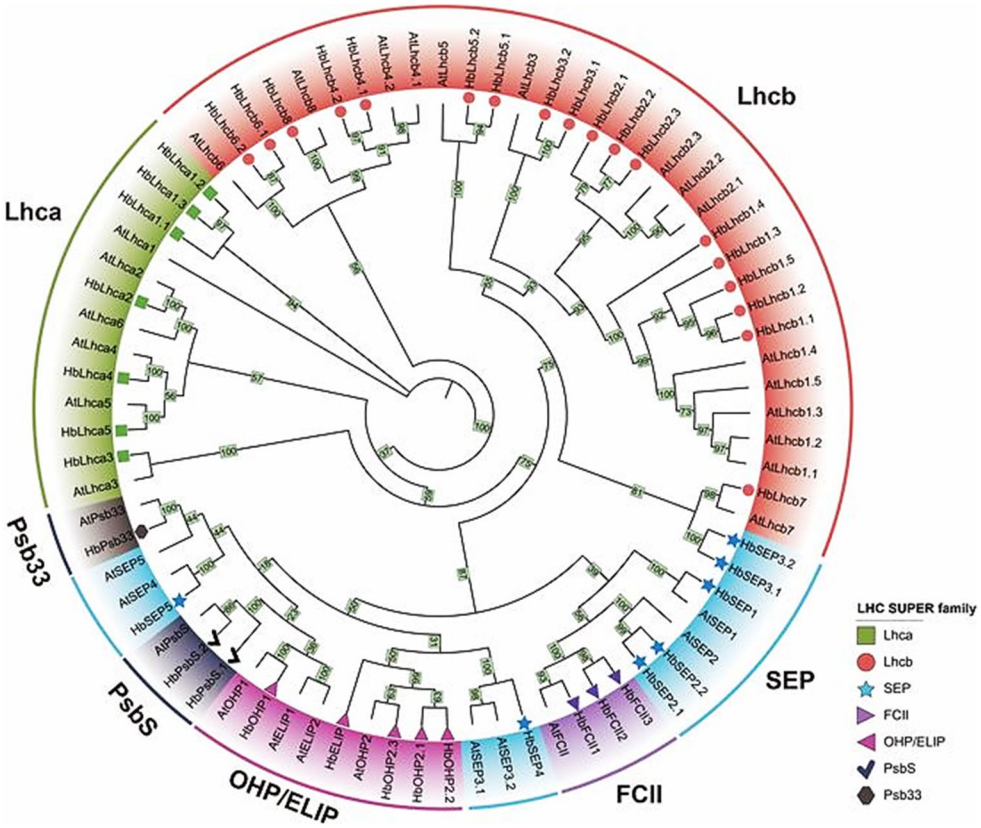

Fig. 3 Maximum-likelihood phylogenetic tree of LHC superfamily proteins from ***H. brasiliensis*** and ***A. thaliana***. Phylogenomic reconstruction resolved six subfamilies (*Lhca*, *Lhcb*, SEP, FCII, OHP, PsbS), with *Lhcb* (18 members) and *Lhca* (7 members) dominating. Bootstrap values ≥ 70% are shown. Branch colors correspond to subfamily classifications. The tree topology reflects lineage-specific expansion events, with tandem duplications driving functional conservation

Table 2 Collinear gene pairs between *H. brasiliensis*, *O. sativa* and *A. thaliana*

Gene1Name	Gene1ID	Gene2ID	Gene2Name
AtSEP2	*NM_127766.3*	GT020542	*HbLhcb1.1*
AtCAB3	*NM_102731.3*	GT029295	*HbLhcb1.4*
AtLHCB3	*NM_124807.4*	GT025912	*HbLhcb3.1*
pre_OsLHC	*XM_015791053.2*	GT009757	*HbLhcb3.2*
pre_OsLHC	*XM_015791053.2*	GT025912	*HbLhcb3.1*
AtLHCB4.1	*NM_120231.4*	GT008959	*HbLhcb4.1*
AtLHCB4.3	*NM_129568.3*	GT038427	*HbLhcb8*
AtYPZB1	*NM_115447.2*	GT038427	*HbLhcb8*
AtYPZB2	*NM_001084666.2*	GT038427	*HbLhcb8*
AtOHP	*NM_120290.3*	GT015156	*HbOHP1*
AtOHP2	*NM_103122.4*	GT020618	*HbOHP2.2*
AtOHP2	*NM_103122.4*	GT020750	*HbOHP2.3*
AtNPQ4	*NM_001333231.1*	GT002047	*HbPsbS.1*
AtELIP1	*NM_113183.4*	GT021010	*HbSEP2.1*
AtELIP2	*NM_117551.3*	GT021010	*HbSEP2.1*
OsJGP	*XM_015757489.2*	GT021010	*HbSEP2.1*
AtLIL3.1	*NM_117868.6*	GT039527	*HbSEP4*
AtLIL3.2	*NM_124082.5*	GT039527	*HbSEP4*

Phylogenomic reconstruction and syntenic conservation

Phylogenetic analysis of *H. brasiliensis* and *A. thaliana* LHC proteins resolved six subfamilies (SEP, FCII, OHP, PsbS, *Lhca*, *Lhcb*), with uneven distribution dominated by *Lhcb* (18 members) and *Lhca* (7 members), while SEP, FCII, OHP, and PsbS comprised 2–4 genes each (Fig. 3). Synteny analysis identified tandem duplication-driven family expansion, evidenced by intra-chromosomal clusters of *HbLHC* genes. Selection pressure analysis revealed strong purifying selection ($Ka/Ks < 1$) in 13/14 paralog pairs (e.g., *HbOHP2.3/HbOHP2.1*), with positive selection detected in *HbOHP2.2/HbOHP2.3* ($Ka/Ks = 1.32$), reflecting evolutionary conservation of photosynthetic core functions (Table 2). Cross-species comparisons showed stronger syntenic conservation with *A. thaliana* (15 orthologs, including *HbLhcb1.4/AtCAB3*) than with *O. sativa* (3 pairs), indicating dicot-specific functional conservation. Notably, *HbLhcb8* exhibited expanded orthology to three *A. thaliana* loci (*AtLHCB4.3*, *AtYPZB1/2*), while *HbSEP2.1* associated with multiple *A. thaliana* ELIP genes (*AtELIP1/2*, *AtJGP*), suggesting post-divergence neofunctionalization (Fig. 4). These patterns highlight conserved architectural frameworks alongside lineage-specific innovations in LHC family evolution.

Cis-regulatory element landscapes in *HbLHC* promoters

Cis-regulatory analysis of *HbLHC* promoters identified stress-responsive (MYB/MYC), hormone-regulated

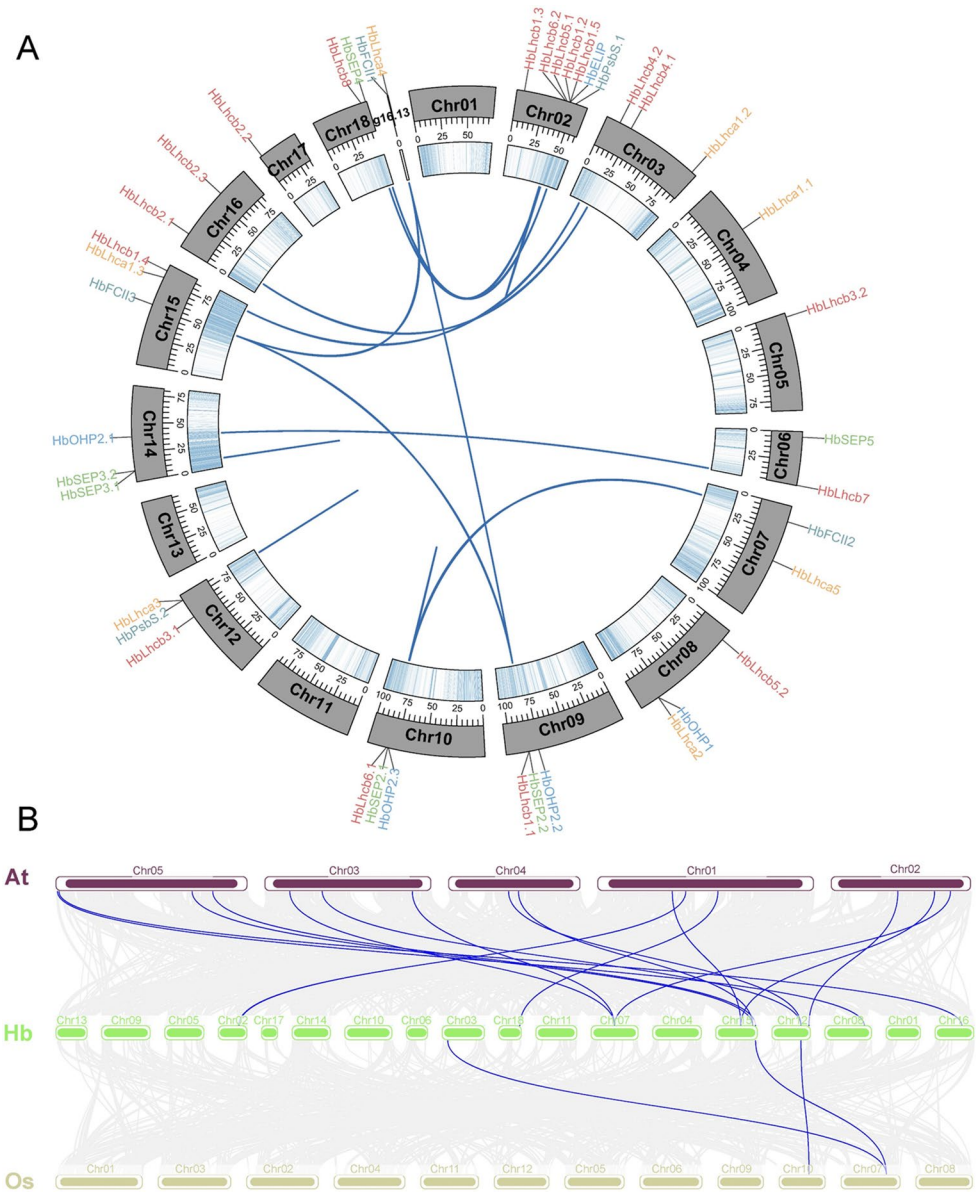

Fig. 4 Comparative synteny analysis of ***LHC*** superfamily genes in ***H. brasiliensis*** and related species. (**A**) Synteny dot plot illustrating orthologous gene pairs between *H. brasiliensis* and *A. thaliana*, with 15 identified orthologs including the highlighted *HbLhcb1.4*/*AtCAB3*. (**B**) *H. brasiliensis* vs. *O. sativa*: Limited conservation (3 orthologs). Expanded orthology events highlight lineage-specific innovations: *HbLhcb8* links to three *A. thaliana* loci (*AtLHCB4.3*, *AtYPZB1/2*); *HbSEP2.1* associates with *A. thaliana* ELIP genes (*AtELIP1/2*, *AtJGP*), suggesting post-divergence neofunctionalization

(ABRE/ERE), and growth-related (Box4/G-box) elements. MYB-binding sites (CCAAT-box) were enriched in *Lhca4* (12 sites), implicating photosynthetic plasticity to environmental fluctuations. MYC motifs (CANNTG) peaked in *Lhca3* and *Lhcb2.2* (11 sites each), potentially mediating cold adaptation via ICE1-CBF-like pathways. Stress-specialized loci included *Lhcb8* (11 AREs for hypoxia response) and *Lhcb1.4* (11 STREs for oxidative/thermal stress), co-harboring biotic defense elements (W-box/WUN-motif).

Hormone regulation showed subfamily divergence: ABA-responsive *Lhca1.3* (8 ABREs) mirrored *AtLhcb2* cold sensitivity, while ethylene signaling dominated *FCII1* (11 EREs). Photoperiodic control emerged through light-responsive Box4 (8.3–70% frequency) and G-box (50% in *SEP2.1*). The omnipresent GT1-motif suggested auxiliary salt-stress roles, particularly in *Lhca3*—a regulatory nexus integrating environmental sensing with photoprotection (Fig. 5). These cis-element architectures reflect evolutionary trade-offs between conserved stress responses and lineage-specific adaptations in *H. brasiliensis*.

Protein-protein interaction networks of the LHC protein family in *Hevea brasiliensis*

Protein interaction network analysis of 57 *H. brasiliensis* LHC components—including photosystem I (PSI), photosystem II (PSII), and auxiliary proteins—demonstrated a hierarchical photosynthetic architecture (Fig. 6). PSI/PSII subunits formed densely connected hubs with centrality metrics (degree > 15) confirming their regulatory

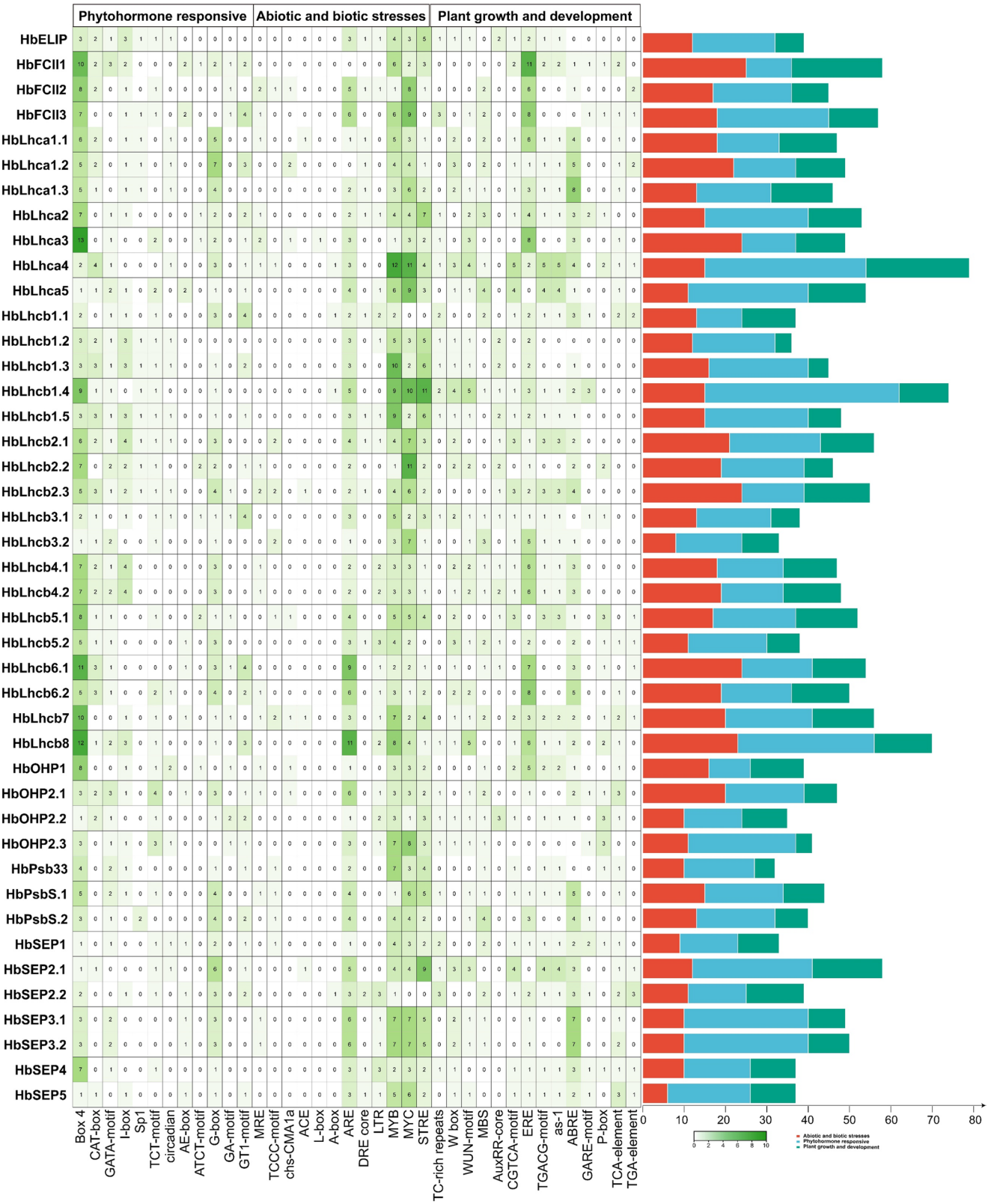

Fig. 5 Cis-regulatory element analysis of ***H. brasiliensis LHC*** gene promoters**.** Promoter regions (2 kb upstream) were analyzed using PlantCARE. Enriched elements include MYB/MYC (stress response), ABRE/ERE (hormone signaling), and Box4/G-box (light regulation). *HbLhca3* and *HbLhcb2.2* have high MYC motif counts (11 sites each); *HbLhca4* contains 12 MYB-binding sites. Color gradient represents motif frequency per promoter

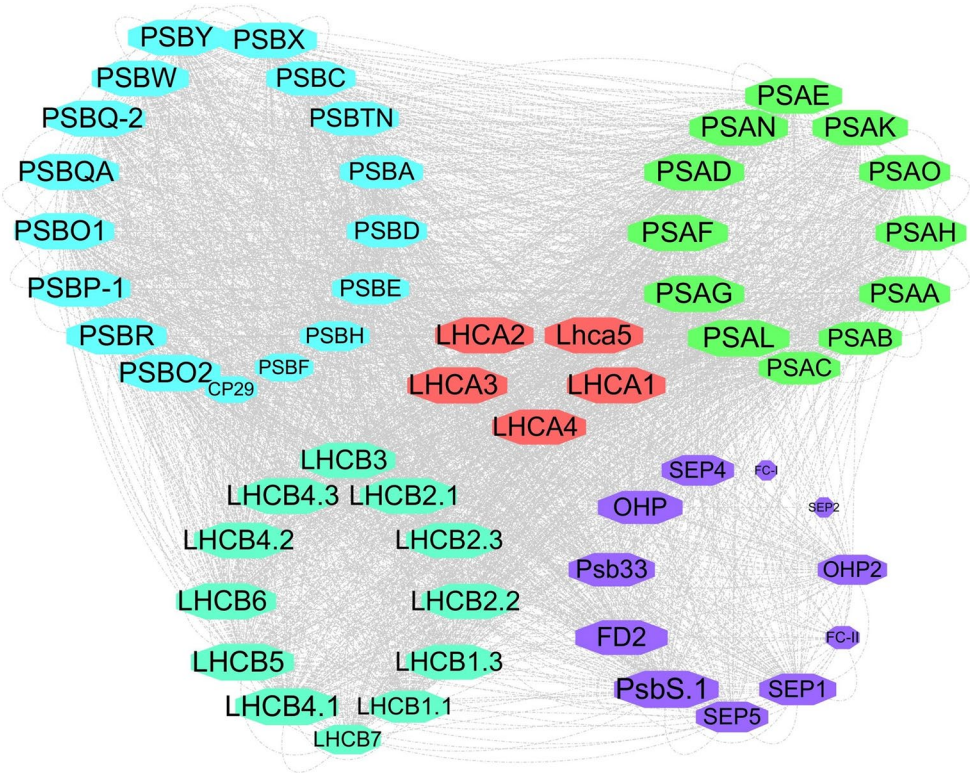

Fig. 6 PPI network analysis of photosystem-associated proteins. PPI networks were constructed using STRING v12.0and analyzed in Cytoscape v3.9.1. Hub proteins (degree > 15) include PSI/PSII core subunits. *HbLhca1.3* and *HbLhcb4.1* mediate PSI-PSII connectivity (red edges). Peripheral accessory proteins (gray nodes) show stress-inducible expression (83%)

dominance. This topology correlated with leaf-specific transcriptional abundance, reflecting functional prioritization of light-harvesting machinery. *HbLhca1.3* and *HbLhcb4.1* emerged as critical bridges mediating PSI-PSII synergy, while accessory proteins maintained selective connectivity to core complexes despite peripheral localization, suggesting specialized roles in stress-responsive photosynthetic tuning. The network's high redundancy aligns with evolutionarily conserved robustness in energy transduction, where hub redundancy buffers environmental perturbations. Strikingly, 83% of peripheral accessory proteins exhibited stress-inducible expression patterns, implicating their adaptive roles in maintaining efficiency under suboptimal conditions.

Structural conservation and functional specialization of *HbLHC* proteins

Structural characterization of *H. brasiliensis* LHC proteins revealed conserved α-helical folds in *Lhca1-5* and β-sheet/α-helix alternations in *Lhcb1-8*, with RMSD < 1.5 Å against *A. thaliana* homologs (Fig. 7). Subfamilies exhibited compartment-specific topologies: *Lhca* adopted PSI-optimized conformations, while *Lhcb* displayed PSII-associated architectures. Ultraconserved chlorophyll-binding domains were identified, including histidine-glutamine clusters (*Lhca*: His87/Gln102; *Lhcb*: His83/Gln98) critical for photon capture. Cross-species alignment revealed 92–97% identity at chlorophyll-binding sites, with strong purifying selection ($dN/dS < 0.25$) underscoring evolutionary constraint on light-harvesting efficiency. Lineage-specific loop modifications in *HbLhcb6* and *HbLhca3* (RMSD > 2.0 Å vs. monocots) suggested neofunctionalization in shade adaptation, contrasting with global structural conservation. These findings position *HbLHCs* as evolutionarily refined photonic units, where peripheral domain plasticity enables biotechnological optimization of quantum yield without core functional compromise.

Spatiotemporal expression profiling of *HbLHC* transcripts

Tissue-specific expression profiling of *HbLHC* genes across *H. brasiliensis* organs revealed functional diversification aligned with physiological demands (Fig. 8). Photosynthetic dominance was evident, with 68% of *HbLHCs* exhibiting leaf-specific expression (e.g., *Lhcb3.1*: FPKM 38.2). Reproductive tissues showed specialized regulation: *Lhca3* displayed 4.7-fold enrichment in female flowers, while *Lhcb3.1* maintained dual expression peaks in leaves and male flowers (FPKM 29.4), suggesting roles in gametophyte development. Non-photosynthetic niches included latex-specific *ELIP* overexpression (FPKM 15.8 vs. <2.0 in other tissues), potentially countering oxidative stress

during latex biosynthesis, and seed-enriched *OHP1* (FPKM 12.3), implicating desiccation tolerance. Constitutively low-expressed genes (*Lhcb1.4*, *FCII3*; FPKM < 1.5) indicated functional redundancy, whereas uniform low expression of *Lhca4/FCII1* (FPKM 2.1 ± 0.3) despite divergent promoters suggested post-transcriptional control. These expression dichotomies—photosynthetic prioritization versus tissue-specific adaptation—reflect evolutionary optimization balancing light-harvesting efficiency with lineage-specific innovations in latex production and tropical reproduction.

Cold stress-responsive dynamics of *HbLHC* genes and photosystem components

Quantitative RT-PCR analysis of *HbLHC* superfamily members under progressive low-temperature stress (LL1–LL7: 18 °C/14°C day/night for 1, 3, 5, and 7 days; Fig. 9A) revealed stage-specific transcriptional reprogramming in *H. brasiliensis* leaves. A majority of *HbLHC* genes exhibited downregulation, with distinct temporal patterns among subfamilies. For instance, *HbLhcb6* and *HbOHP2* showed biphasic responses—initial suppression during LL1–LL3, partial transcriptional recovery at LL5 ($p < 0.05$, ANOVA with Tukey's test), followed by secondary repression at LL7. Conversely, *HbPsb28* expression progressively increased, reaching a 2.3-fold induction at LL7 ($p < 0.01$).

Core photosystem I components (*HbLhca2*, *HbLhca4*) underwent sustained repression, with transcript levels declining over fourfold at LL5 and partially recovering to 1.8-fold relative to LL5 at LL7, reflecting a delayed chloroplast acclimation process. The *HbFCII* subfamily exhibited temporally divergent expression: *HbFCII1* showed delayed but robust induction (7.1-fold at LL7 vs. LL1; $p < 0.001$), whereas *HbFCII2* peaked transiently at LL3 (3.4-fold; $p < 0.05$), and *HbFCII3* responded acutely at LL1 (5.2-fold; $p < 0.01$), indicating subfunctional divergence within the subfamily.

In contrast, *HbSEP3* expression declined steadily throughout cold exposure, reaching 0.15-fold at LL7 ($p < 0.001$), a trend associated with progressive photosystem destabilization. Genes such as *HbLhcb3.1* and *HbELIP* displayed constitutively low and non-responsive expression (< 0.5-fold change; $p > 0.05$), while *HbOHP2* oscillated between suppression and transient recovery (LL1: 0.3-fold; LL5: 1.1-fold; LL7: 0.2-fold; $p < 0.05$ at LL5).

Collectively, these transcriptional patterns delineate a triphasic cold-response model: (1) immediate suppression of light-harvesting components (LL1–LL3), (2) transient photoprotection via partial gene reactivation (LL5), and (3) systemic acclimation and reprogramming of photosystem components (LL7).

At the protein level, subcellular localization assays confirmed chloroplast accumulation of HbLhcb2.2 and cytoplasmic localization of HbELIP under cold conditions (Fig. 9B), supporting functional neofunctionalization beyond canonical chloroplast targeting predicted in silico.

Discussion

The light-harvesting chlorophyll a/b-binding (LHC) gene family is central to regulating photosynthetic efficiency and enhancing resilience to abiotic stresses. While extensively characterized in model plants such as *Arabidopsis thaliana* and major crops including *O. sativa* and *Z. mays* (Jansson 1994; Rochaix and Bassi 2019), LHC family members in *H. brasiliensis*—a globally important latex-producing species—remain largely unstudied. In this study, we report the first genome-wide identification and characterization of 43 *HbLHC* genes, classified into six conserved subfamilies. Despite retaining canonical chlorophyll-binding domains homologous to *A. thaliana* (Gu et al. 2017), these genes display distinct transcriptional responses to cold stress, likely reflecting an evolutionary trade-off between photosynthetic efficiency and latex metabolism in laticiferous tissues (Gong et al. 2018).

Phylogenomic analysis revealed strict conservation of key residues required for photon capture (e.g., His183 for chlorophyll ligation), underscoring evolutionary constraints on photosystem functionality (Woodrow et al. 2012). However, in contrast to the cold-inducible *AtLhcb1* paradigm (Pietrzykowska et al. 2014), transcriptional profiling under cold stress showed significant suppression of multiple *HbLhca/b* homologs ($\log_2 FC \leq -2.0$, $p < 0.01$). This divergence may reflect a species-specific metabolic reprogramming strategy, wherein downregulation of light-harvesting genes reallocates resources toward latex biosynthesis—a survival mechanism aimed at preserving economic productivity under suboptimal temperatures. Similar trade-offs have been reported in *Populus*, where perennial growth habits favor resource allocation plasticity under stress (Zhao et al. 2020).

Notably, our protein–protein interaction (PPI) network analysis demonstrated that hub proteins such as HbLhca1.3 and HbLhcb4.1 mediate critical connectivity between PSI and PSII supercomplexes, while 83% of peripheral accessory proteins exhibited stress-inducible expression. This hierarchical and redundant architecture likely underpins the robustness of photosynthetic integrity under chilling conditions, paralleling the modular photoprotective networks described in *A. thaliana* (Müller et al. 2021; Szklarczyk et al. 2023). The enrichment of stress-responsive *cis*-elements

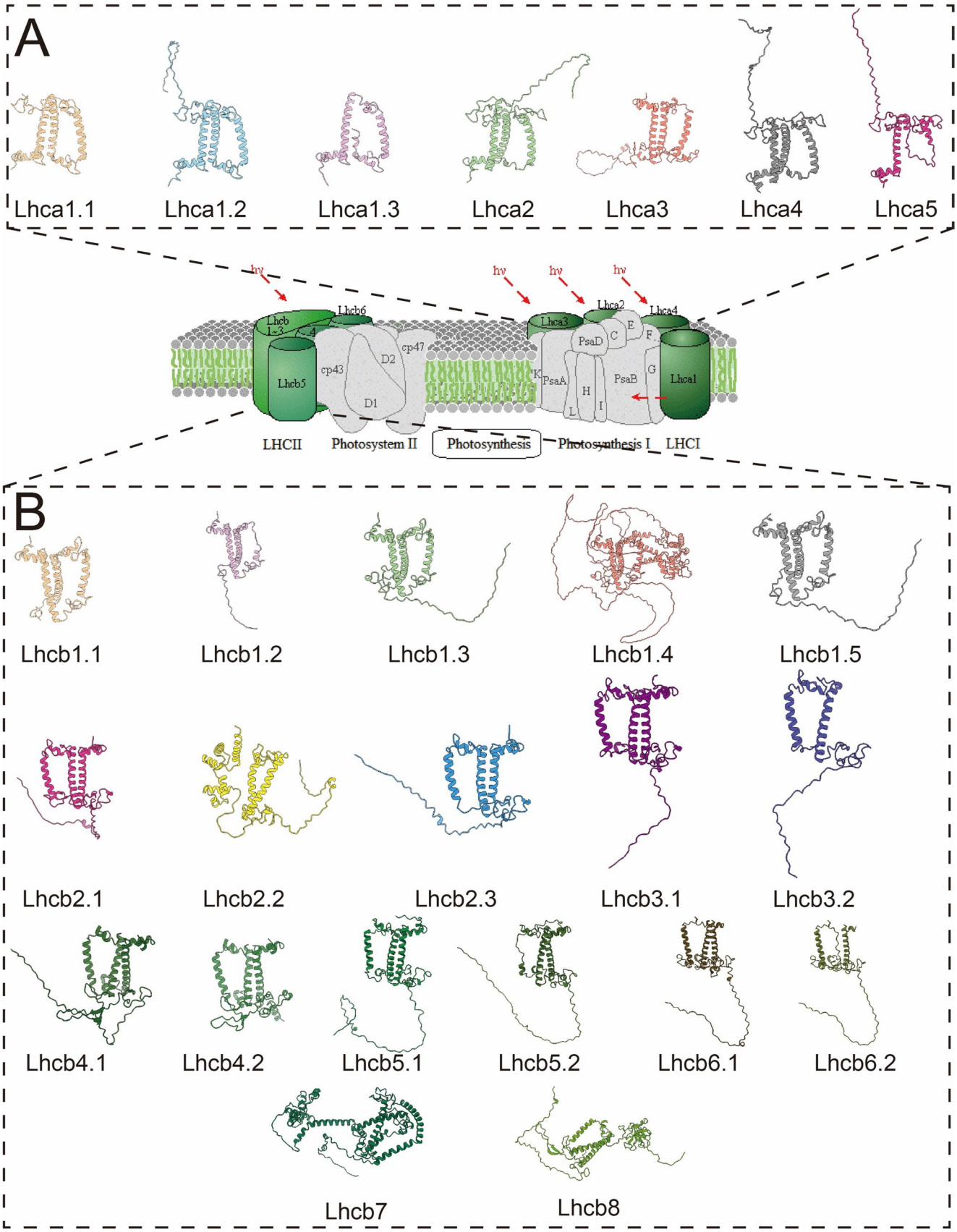
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Fig. 7 Structural modeling of LHC proteins in ***H. brasiliensis***. Predicted structures of Lhca1–5 and Lhcb1–8 show conserved α-helical (Lhca) and mixed β-sheet/α-helix (Lhcb) folds, closely matching *A. thaliana* homologs (RMSD < 1.5 Å). PSI- and PSII-associated proteins exhibit compartment-specific topologies. Chlorophyll-binding domains contain conserved His/Gln residues (Lhca: His87/Gln102; Lhcb: His83/Gln98), with 92–97% identity across species and strong purifying selection (*dN/dS* < 0.25). *HbLhcb6* and *HbLhca3* display lineage-specific loop variation (RMSD > 2.0 Å vs. monocots), suggesting potential neofunctionalization

(e.g., MYB/MYC) in promoters of cold-repressed *HbLHCs* further supports transcriptional regulation via stress-responsive TFs, potentially orchestrating a switch between photoprotection and secondary metabolism (Gong et al. 2018; Chen et al. 2022).

Gene duplication analyses further suggest that tandem duplication events have contributed to the expansion and potential functional redundancy of the *HbLHC* family, particularly under high-light tropical environments (Liu et al. 2023). Notably, our experimental subcellular localization assays revealed previously unrecognized protein targeting. Although *HbLhcb2.2* was predicted to localize to mitochondria, fluorescent protein-tagging demonstrated chloroplast localization under cold stress (Fig. 9B), consistent with its role in photosystem II. Conversely, HbELIP localized to the cytoplasm in laticifer cells, implying functional divergence. These findings provide insight into the paradox of extra-chloroplastic LHC proteins retaining chlorophyll-binding domains. We propose that HbLhcb2.2 may still interact with tetrapyrrole intermediates via conserved His/Gln residues (Fig. 7), while HbELIP may repurpose its hydrophobic regions for participation in cytosolic antioxidant complexes, possibly contributing to latex oxidative stress mitigation.

Beyond transcriptional control, emerging evidence suggests epigenetic regulation plays a role in *H. brasiliensis* cold adaptation. For instance, chilling-induced DNA demethylation has been associated with cold tolerance in rubber trees (Tang et al. 2022). Such epigenetic memory could underlie the delayed induction of *HbFCII1* and the sustained repression of *HbSEP3* observed here, echoing similar mechanisms in perennial crops (Müller et al. 2021).

Among the candidate genes identified, *HbPsbS*, *HbFCII1*, and *HbLhcb2.2* emerge as promising targets for genetic improvement. *HbPsbS* ($\log_2$ FC = 1.8 at LL7) likely facilitates non-photochemical quenching (NPQ) to dissipate excess energy, as observed in *A. thaliana* (Niyogi et al. 2021), while *HbFCII1*'s delayed induction (7.1-fold at LL7) suggests a role in late-stage photoprotection or chaperone activity. Conversely, *HbLhcb1.4* (constitutively low-expressed) may be a candidate for knockdown to reduce resource diversion from latex production under cold stress. We propose a modular engineering strategy: overexpressing *HbPsbS* and *HbFCII1* while silencing cold-sensitive genes like *HbLhcb1.4* to enhance cold resilience without compromising yield. This approach aligns with successful efforts in apple, where *MdLhcb4.3* overexpression conferred dual tolerance to chilling and osmotic stress (Zhao et al. 2020).

Furthermore, the tissue-specific expression of *HbLHCs* (e.g., *HbLhca3* in reproductive tissues and *HbELIP* in laticifers) underscores functional diversification beyond photosynthesis. The cytoplasmic localization of HbELIP implies neo-functionalization in oxidative stress mitigation during latex biosynthesis, a hypothesis supported by its orthology to *A. thaliana* ELIPs involved in ROS scavenging (Shimizu et al. 2020). Similarly, the seed-enriched expression of *HbOHP1* suggests a role in desiccation tolerance, as reported for orthologs in *O. sativa* (Zhang et al. 2022).

Interestingly, synteny analysis revealed stronger conservation with *A. thaliana* than *O. sativa*, indicating dicot-specific evolutionary trajectories. For example, the expansion of *HbLhcb8* orthology to multiple *A. thaliana* loci (*AtLHCB4.3*, *AtYPZB1/2*) and the association of *HbSEP2.1* with ELIP genes suggest post-divergence neofunctionalization potentially linked to laticifer metabolism in *Hevea* (Tang et al. 2016; Liu et al. 2023).

To translate these findings, future work should prioritize: (i) CRISPR/Cas9-based functional validation of *HbPsbS* and *HbFCII1* in transgenic *Hevea*; (ii) haplotype analysis of natural *HbLHC* alleles in wild germplasm (e.g., cold-tolerant accessions from Rodonia; Mao et al. 2023) to identify breeding markers; and (iii) structural studies using cryo-EM to resolve HbLHC-photosystem complexes (Ravi and Tan 2015). Additionally, integrative multi-omics approaches, such as combining metabolomics with transcriptomics to link *HbLHC* expression with flavonoid and arginine pathways (Mao et al. 2023), may provide a systems-level view of cold stress adaptation.

Promoter analysis showed enrichment of MYB-binding motifs among cold-repressed *HbLHC* genes (Gong et al. 2018), indicating transcriptional regulation by stress-responsive TFs. This supports the existence of a transcriptional "switch" modulating the balance between photoprotection and secondary metabolism. Nevertheless, certain limitations should be acknowledged. First, protein abundance under cold stress was not quantified by immunoblotting, leaving post-transcriptional dynamics unresolved. Second, the experimental cold treatments were conducted under controlled conditions, lacking combinatorial field stressors such as high light or drought.

In conclusion, this study provides a comprehensive molecular framework for understanding *HbLHC*-mediated cold adaptation in *H. brasiliensis*. By identifying *HbPsbS* and *HbFCII1* as positive regulators and *HbLhcb1.4* as a likely negative regulator, and by proposing specific manipulation strategies, we present a concrete roadmap for breeding cold-resilient rubber

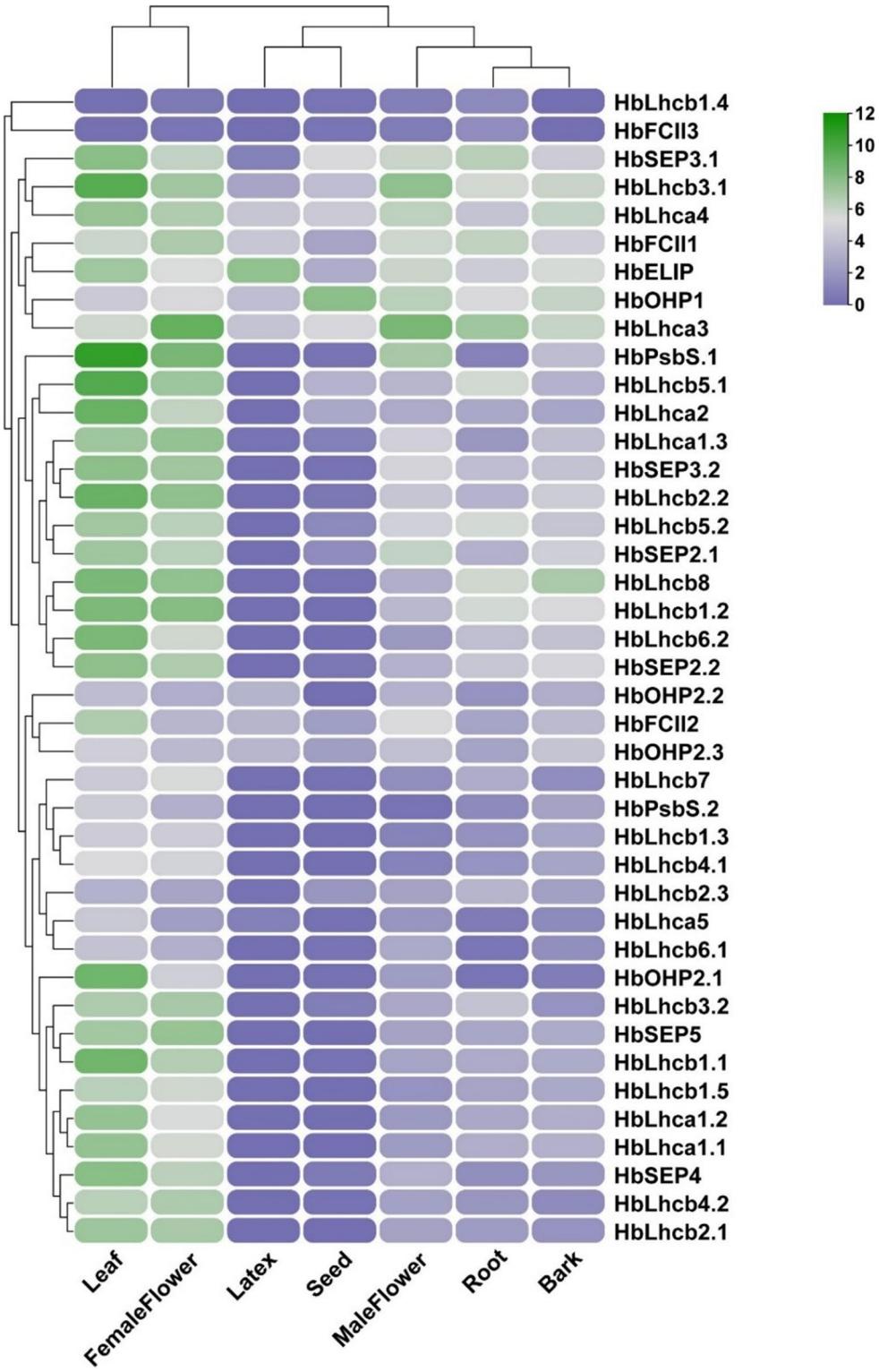

Fig. 8 Tissue-specific expression of ***HbLHC*** genes in ***H. brasiliensis***. Hierarchical clustering of $\log_2$ (TPM + 1)-normalized expression across leaf, latex, flower, and seed indicates functional divergence. 68% of *HbLHC* genes show leaf-preferential expression (e.g., *HbLhcb3.1*, FPKM 38.2). Reproductive specificity includes *HbLhca3* (4.7-fold enrichment in female flowers) and dual-peaked *HbLhcb3.1* (leaf and male flowers, FPKM 29.4). Non-photosynthetic expression includes latex-specific HbELIP (FPKM 15.8 vs. <2.0) and seed-enriched HbOHP1 (FPKM 12.3), suggesting roles in stress tolerance. Low expression of *HbLhcb1.4* and *HbFCII3* (FPKM < 1.5) indicates possible redundancy; *HbLhca4* and *HbFCII1* show uniform low expression (FPKM 2.1 ± 0.3), implying post-transcriptional regulation

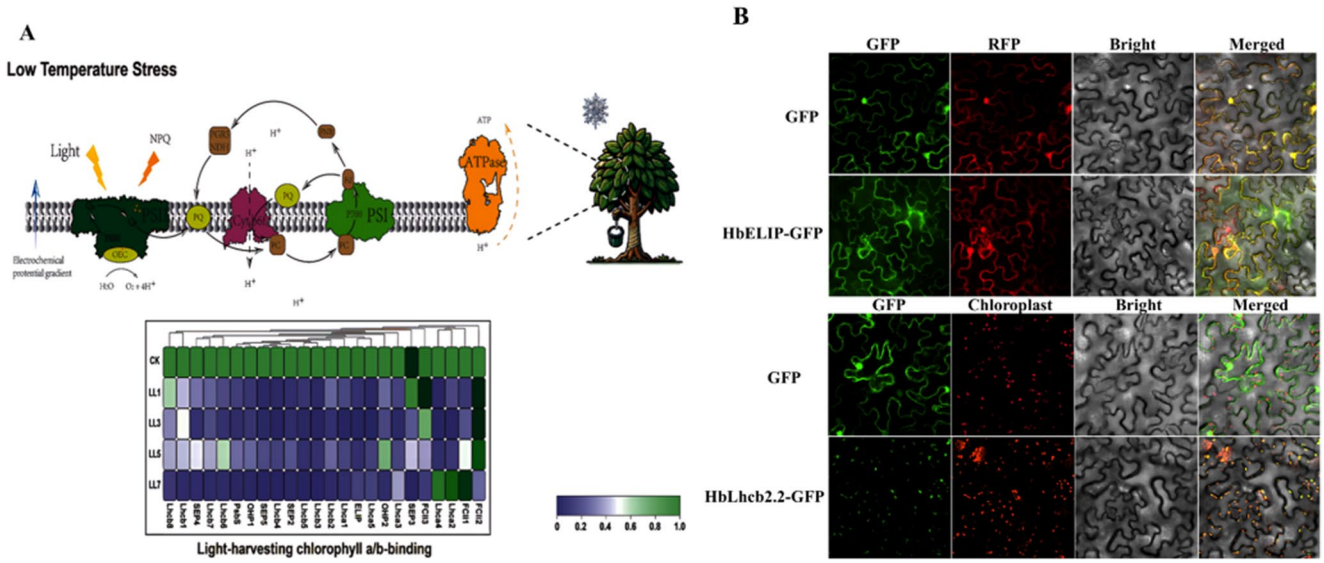

Fig. 9 Cold-responsive expression and subcellular localization of LHC genes in ***H. brasiliensis***. (**A**) qRT-PCR analysis of LHC gene expression under progressive low-temperature stress (LL1–LL7; 18 °C/14°C day/night). Expression profiles show triphasic regulation: early suppression (LL1–LL3), partial recovery (LL5), and systemic acclimation (LL7). *HbPsb28* peaked at 2.3-fold at LL7 ($p<0.01$); *FCII1* showed delayed induction (7.1-fold). Error bars indicate SEM ($n=3$ biological replicates). (**B**) Subcellular localization of HbLhcb2.2 and HbELIP in *N. benthamiana* leaves. Despite its predicted mitochondrial targeting, HbLhcb2.2 localized to the chloroplast; HbELIP was observed in the cytoplasm. Scale bar = 20 µm

cultivars. Our findings highlight how evolutionary diversification, transcriptional plasticity, and subcellular neofunctionalization contribute to balancing photosynthesis and stress resilience in an economically vital perennial crop.Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11295-025-01713-w.

Author contributions XC W designed the research and made major revisions to the manuscript. AF L wrote the initial draft of the manuscript. AF L and Y Y organized the data and visualized the images. SL T conducted the experiments. Y Y, SL T, FY F, M W, YX L, and MW Y contributed to the editing of the manuscript.

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Data availability All genomic and transcriptomic datasets generated in this study have been deposited in the National Genomics Data Center (NGDC) under BioProject accession number PRJNA310171. Processed data and analysis scripts are available upon request from the corresponding author.

Declarations

Conflict of interest The authors declare no competing financial interests or personal relationships that could influence the work reported in this study.

Data Archiving Statement All genomic and transcriptomic data analyzed in this study were obtained from publicly available databases. Specifically, the genome assembly and annotated proteome of *Hevea brasiliensis* (accession: GCA_018446215.1) were retrieved from the National Genomics Data Center (NGDC, https://ngdc.cncb.ac.cn/). The RNA-seq datasets used for expression analysis, including tissue-specific and cold stress conditions, were downloaded from the NCBI Sequence Read Archive (SRA) under BioProject accession number PRJNA310171. No new raw sequencing data were generated in this study. All processed data and scripts used for analysis are available from the corresponding author upon reasonable request.

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